

Week 4

1. Gradients of Functions Expressed as a Multiplication of Matrices	1
2. Linear Regression	3
3. Least Squares	7
4. Maximum Likelihood	8
5. Probabilistic View of Linear Regression	10
6. Polynomial Regression	12
7. Eigenvalues & Eigenvectors	14
8. Properties of Eigenvalues and Eigenvectors	18
9. Similar Matrices	24
10. Diagonalization	25
11. Fibonacci Sequence	28
12. Symmetric Matrices and Orthogonal Matrices	31
13. Eigenvalues of Triangular Matrices and Projection Matrices	32
14. Ridge Regression	34

1. Gradients of Functions Expressed as a Multiplication of Matrices

Consider the following vectors and matrices.

$$x = \begin{bmatrix} p \\ q \end{bmatrix}, y = \begin{bmatrix} m \\ n \end{bmatrix}, S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

- S is a symmetric matrix.
- $a, b, c, m, n \in \mathbb{R}$ are constants
- p, q are the parameters of the functions f and g .

$$f(p, q) = x^T S x$$

$$g(p, q) = x^T y$$

$$h(p, q) = x^T x$$

Goal : Find:

- $\nabla f(p, q)$
- $\nabla g(p, q)$
- $\nabla h(p, q)$

Finding these two will be useful for linear regression with squared error.

Let us expand $f(p, q)$

$$\begin{aligned}f(p, q) &= x^T S x \\&= \begin{bmatrix} p & q \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} \\&= \begin{bmatrix} p & q \end{bmatrix} \begin{bmatrix} ap + bq \\ bp + cq \end{bmatrix} \\&= ap^2 + bpq + bpq + cq^2 \\&= ap^2 + 2bpq + cq^2\end{aligned}$$

$\nabla f(p, q)$ is given as

$$\begin{aligned}\nabla f(p, q) &= \begin{bmatrix} 2ap + 2bq \\ 2bp + 2cq \end{bmatrix} \\&= 2 \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} \\&= 2Sx\end{aligned}$$

Let us find $g(p, q)$

$$\begin{aligned}g(p, q) &= x^T y \\&= pm + qn\end{aligned}$$

$\nabla g(p, q)$ is given as

$$\begin{aligned}\nabla g(p, q) &= \begin{bmatrix} m \\ n \end{bmatrix} \\&= y\end{aligned}$$

Let us find $h(p, q)$

$$h(p, q) = x^T x$$

$$= p^2 + q^2$$

$\nabla h(p, q)$ is given by

$$\begin{aligned}\nabla h(p, q) &= \begin{bmatrix} 2p \\ 2q \end{bmatrix} \\ &= 2x\end{aligned}$$

These three results are true for any $n \times n$ symmetric matrix S , any $n \times 1$ vectors x and y , where $n \geq 1$.

2. Linear Regression

Consider the following dataset:

$$D = \{(x_1, y_1), \dots, (x_n, y_n)\}$$

where, $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$, for $i \in \{1, \dots, n\}$

Each data point has d features. The i^{th} data point is given as

$$x_i = (x_{(i,1)}, \dots, x_{(i,d)})$$

The corresponding label is y_i .

Assumption

In linear regression, we assume that the data points and labels are linearly related. That is:

$$\begin{aligned}\mathbf{w}_0 + \mathbf{w}_1 x_{(1,1)} + \dots + \mathbf{w}_d x_{(1,d)} &= y_1 \\ \vdots &\vdots \\ \mathbf{w}_0 + \mathbf{w}_1 x_{(n,1)} + \dots + \mathbf{w}_d x_{(n,d)} &= y_n\end{aligned}$$

Here, $\mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_n$ are unknown. The matrix form of the above system of linear equations is $X\mathbf{w} = y$, where,

$$X = \begin{bmatrix} 1 & x_{(1,1)} & \cdots & x_{(1,d)} \\ 1 & x_{(2,1)} & \cdots & x_{(2,d)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{(n,1)} & \cdots & x_{(n,d)} \end{bmatrix}_{n \times (d+1)}, \quad \mathbf{w} = \begin{bmatrix} \mathbf{w}_0 \\ \mathbf{w}_1 \\ \vdots \\ \mathbf{w}_d \end{bmatrix}_{(d+1) \times 1}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}$$

Each row in X corresponds to a data point, and each column in X corresponds to a feature. $\mathbf{w}_0$ captures

the unknown constant term in a linear equation. Every entry in the first column of X is 1. This is done to incorporate the unknown constant term $\mathbf{w}_0$ into the matrix multiplication $X\mathbf{w}$. The first column is also referred to as the **dummy** feature column.

Goal

Using the training dataset D , learn a linear function $f: \mathbb{R}^{d+1} \rightarrow \mathbb{R}$, which takes a data point as input and outputs a prediction for its label. That is, the goal is to learn the unknown vector $\mathbf{w}$ and predict the labels of the unknown data point. If $\hat{x} \in \mathbb{R}^{d+1}$ is a test data point, then the prediction for the label of $\hat{x}$ is given by:

$$\begin{aligned} f(\hat{x}) &= \hat{x}^T \mathbf{w} \\ &= \mathbf{w}_0 \cdot 1 + \mathbf{w}_1 \cdot \hat{x}_1 + \dots + \mathbf{w}_d \cdot \hat{x}_d \end{aligned}$$

Training Error

The training error is a measure of the inaccuracies in the prediction of the labels of the training data points. A good prediction function $f(x)$ will be learnt by minimizing the error. There are many ways to define the training error. Here are two intuitive ways to define the training error.

Sum Absolute Error (SAE)

The sum absolute error is given by

$$\text{SAE} = \sum_{i=1}^n |f(x_i) - y_i| = \sum_{i=1}^n |x_i^T \mathbf{w} - y_i| = \|X\mathbf{w} - y\|_1$$

SAE is the $\mathbf{l}_1$ norm of the error vector $X\mathbf{w} - y$. Recall that the $\mathbf{l}_1$ norm of a vector is defined as the sum of the absolute values of all its coordinates.

Sum Squared Error (SSE)

The sum squared error is given by

$$\text{SSE} = \sum_{i=1}^n (f(x_i) - y_i)^2 = \sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2 = \|X\mathbf{w} - y\|_2^2$$

SAE is the square of the $\mathbf{l}_2$ norm of the error vector $X\mathbf{w} - y$. Recall that the $\mathbf{l}_2$ norm is the Euclidean norm.

Both **SAE** and **SSE** captures the difference between the prediction and the actual label. While the sum

absolute error is difficult to differentiate, the sum squared error is easy to differentiate. Therefore, we will make use of the **SSE**.

Optimization

The unknown vector $\mathbf{w}$ is given by:

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left\| X\mathbf{w} - y \right\|_2^2$$

Here, $\left\| X\mathbf{w} - y \right\|_2^2$ is our objective function and $\mathbf{w}$ is the the argument that minimizes the objective function. Let us denote our objective function as $g(\mathbf{w})$

$$g(\mathbf{w}) = \left\| X\mathbf{w} - y \right\|_2^2$$

Let us expand the objective function

$$\begin{aligned} g(\mathbf{w}) &= \left\| X\mathbf{w} - y \right\|_2^2 \\ &= (X\mathbf{w} - y)^T (X\mathbf{w} - y) \\ &= \mathbf{w}^T X^T X \mathbf{w} - \mathbf{w}^T X^T y - y^T X \mathbf{w} + y^T y \\ &= \mathbf{w}^T X^T X \mathbf{w} - 2\mathbf{w}^T X^T y + y^T y \end{aligned}$$

Note that $y^T X \mathbf{w} = \mathbf{w}^T X^T y$. Now, $\mathbf{w}$ is given as

$$\mathbf{w} = \arg \min_{\mathbf{w}} [g(\mathbf{w})] = \arg \min_{\mathbf{w}} \left[\mathbf{w}^T X^T X \mathbf{w} - 2\mathbf{w}^T X^T y + y^T y \right]$$

Let us find the gradient of g . We shall make use of the fact that $X^T X$ is a symmetric matrix.

$$\begin{aligned} \nabla g(\mathbf{w}) &= \frac{\delta}{\delta w} (\mathbf{w}^T X^T X \mathbf{w}) - \frac{\delta}{\delta w} (2\mathbf{w}^T X^T y) + \frac{\delta}{\delta w} (y^T y) \\ &= 2X^T X \mathbf{w} - 2X^T y \end{aligned}$$

$$= 2(X^T X \mathbf{w} - X^T y)$$

At $\nabla g(\mathbf{w}) = 0$,

$$\nabla g(\mathbf{w}) = 0$$

$$2(X^T X \mathbf{w} - X^T y) = 0$$

$$X^T X \mathbf{w} = X^T y$$

$$\mathbf{w} = (X^T X)^{-1} X^T y$$

Caution : In general, the columns of X need not be linearly independent. In such cases $(X^T X)$ is not invertible. Especially, if the number of columns in X is greater than the number of rows in X , then the columns are not linearly independent. That is, if $d + 1 > n$, which means, the number of features (including the first column, all of whose entries are 1) is greater than the number of data points, then the columns of X is not linearly independent. In such cases, $(X^T X)^{-1}$ does not exist. When the inverse does not exist, we make use of what is referred to as the **pseudoinverse**. This is given by $(X^T X)^+$. The symbol "+" denotes the pseudoinverse.

However, for our discussion, we shall assume that the columns of X are linearly independent. As a result, $X^T X$ is invertible.

$\mathbf{w} = (X^T X)^{-1} X^T y$ is the only critical point of $g(\mathbf{w})$. It turns out that this critical point is also the global minimum of $g(\mathbf{w})$. For now, we shall accept this as a fact and later, while learning about convex functions in week-9, we shall show that this critical point is indeed the global minimum of $g(\mathbf{w})$

$$\mathbf{w} = \arg \min_{\mathbf{w}} [g(\mathbf{w})] = (X^T X)^{-1} X^T y$$

Prediction of Training Data Points

The prediction vector of the training data points is given by

$$Xw = X(X^T X)^{-1} X^T y$$

Prediction of a Test Data Point

Given an unknown test data point $\hat{x} \in \mathbb{R}^{d+1}$, the prediction of the label of $\hat{x}$ is given by

$$\hat{x}^T \mathbf{w} = \hat{x}^T (X^T X)^{-1} X^T y$$

Projection View Point

We are already familiar with the expression $X(X^T X)^{-1} X^T y$. This is the orthogonal projection of the vector y onto the column space of X . Each column vector corresponds to a feature. The column space of X is also referred to as the **feature space**.

Geometrically, the prediction of the label vector is the orthogonal projection of the label vector onto the feature space. $X\mathbf{w}$, which is the prediction vector, is a linear combination of the columns of X . Intuitively, prediction for the label vector y is the vector in the feature space, that is closest to y .

3. Least Squares

Fundamentally, $X\mathbf{w} = y$ is a system of linear equations. Here, X is the coefficient matrix, $\mathbf{w}$ is the unknown, and y is the vector on the right-hand side of the equation.

If y is not in the column space of X , then the linear equation has no solution. When there is no solution, we can find a vector $\hat{y}$ in the column space of X , that is closest to y . Clearly, $\hat{y}$ will be the projection of y onto the column space of X .

$$\hat{y} = X(X^T X)^{-1} X^T y$$

Since $\hat{y}$ is in the column space of X , it will be a linear combination of the column vectors. Therefore, $\hat{y}$ is given by

$$\hat{y} = X\mathbf{w}$$

where, $\mathbf{w}$ is unknown.

From the discussion on linear regression, it is apparent that if $\hat{y}$ is the projection of y , then $\mathbf{w}$ is given by

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left\| X\mathbf{w} - y \right\|_2^2$$

As we are minimizing the squared error, it is often referred to as the method of **least squares**.

4. Maximum Likelihood

IID Random Variables

Consider a coin, when tossed, shows heads with a probability p and shows tails probability $(1 - p)$.

Suppose, X_1 , X_2 and X_3 are three random variables corresponding to three different tosses of the same coin, one after the other. X_i corresponds to the i^{th} toss, where $i \in \{1, 2, 3\}$. Notice the following.

Having the information about the first toss in no way affects the result of second toss. This means,

$$P(X_2 = \text{Heads} | X_1) = P(X_2 = \text{Heads}) \text{ and } P(X_2 = \text{Tails} | X_1) = P(X_2 = \text{Tails})$$

X_1 and X_2 are independent random variables. Similarly, X_1 , X_2 and X_3 are independent.

The probability of seeing heads and the probability of seeing tails are same for all the three tosses. That is

$$\begin{aligned} P(X_1 = \text{Heads}) &= P(X_2 = \text{Heads}) = P(X_3 = \text{Heads}) = p \\ P(X_1 = \text{Tails}) &= P(X_2 = \text{Tails}) = P(X_3 = \text{Tails}) = (1 - p) \end{aligned}$$

Thus, X_1 , X_2 and X_3 are identically distributed.

Together, we can say that X_1 , X_2 and X_3 are IID (independent and identically distributed) random variables.

Likelihood Function

Given $X_1, X_2, \dots, X_3 \stackrel{iid}{\sim} \text{Bernoulli}$. When a coin is tossed, it shows heads with probability p and tails with probability $(1 - p)$. Here, p is unknown.

Let us denote heads with 1 and tails with 0. Suppose we toss the coin thrice and see heads on first toss, tails on second toss, and heads on third toss.

Let us denote the outcomes of the three tosses with small 'x': $x_1 = 1$, $x_2 = 0$ and $x_3 = 1$. Here, capital 'X' is used for the random variable and small 'x' is used for the samples that have been observed.

Then likelihood function, which takes the unknown parameter p and the observed samples as input, is given by

$$L(p; \{x_1, x_2, x_3\}) = P(X_1 = x_1, X_2 = x_2, X_3 = x_3 | p)$$

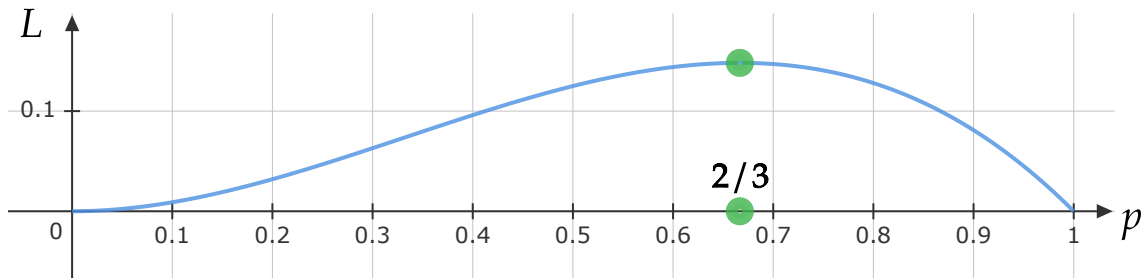
The likelihood function is a joint distribution of the random variables. Since X_1, X_2 and X_3 are IID random variables, we can write the likelihood function as follows:

$$L(p; \{x_1, x_2, x_3\}) = P(X_1 = x_1 | p) \times P(X_2 = x_2 | p) \times P(X_3 = x_3 | p)$$

$P(X_i = 1 | p) = p$ and $P(X_i = 0 | p) = (1 - p)$. We can therefore, write the likelihood function as:

$$L(p; \{x_1, x_2, x_3\}) = p \times (1 - p) \times p = p^2(1 - p)$$

Let us plot the likelihood function.



p is unknown. But, it can take any value from 0 to 1. For different values of p , the likelihood function gives the joint probability of observing $x_1 = 1$, $x_2 = 0$ and $x_3 = 1$.

Since p is unknown, we have to estimate p . One way to estimate p is to find the value of p that maximizes the likelihood function. That is,

$$p = \arg \max_p [L(p; \{x_1, x_2, x_3\})]$$

For the likelihood function plotted above, we can see that it is maximum when $p = \frac{2}{3}$. In our example, $\frac{2}{3}$ is the ratio of the total number of heads and the total number of outcomes.

If the given random variables are assumed to be discrete, then we maximize the joint probability mass function. If the given random variables are assumed to be continuous, we maximize the joint probability density function.

5. Probabilistic View of Linear Regression

Consider the following dataset:

$$D = \{(x_1, y_1), \dots, (x_n, y_n)\}$$

where, $x_i \in \mathbb{R}^{d+1}$ and $y_i \in \mathbb{R}$, for $i \in \{1, \dots, n\}$

Each data point has d features. The i^{th} data point is given as

$$x_i = (1, x_{(i,1)}, \dots, x_{(i,d)})$$

Notice that 1 is the first feature of all the data points. This will be useful to learn the unknown constant term of the linear expression.

The corresponding label is y_i .

Assumption

Suppose we assume that each label y_i , where $i \in \{1, \dots, n\}$ is given by:

$$y_i = x_i^T \mathbf{w} + \epsilon$$

where,

- $\mathbf{w} = (\mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_n)$ is an unknown vector.
- $\epsilon \sim N(0, \sigma^2)$. This is the noise term, with 0 as the expected value.

Each label is assumed to be a linear combination of the features of the corresponding data point along with an added noise whose expected value is 0.

When a constant $c \in \mathbb{R}$ is added to a Gaussian random variable with mean $\mu \in \mathbb{R}$, the result is a Gaussian random variable with mean $\mu + c$. The variance does not change.

Therefore,

$$y_i | x_i = N(x_i^T \mathbf{w}, \sigma^2)$$

We are essentially assuming that for each data point x_i , its corresponding label y_i is a random variable that follows a Gaussian distribution with mean $x_i^T \mathbf{w}$.

Maximum Likelihood

The following is the likelihood function.

$$L(w; D) = \prod_{i=1}^n f(y_i | x_i, w)$$

Notation : $\prod_{i=1}^n i = (1 \times \dots \times n)$. This notation is used for writing products in a compact way. This is similar to the symbol $\sum$, which is used for summation.

Let us expand the likelihood function:

$$\begin{aligned} L(w; D) &= \prod_{i=1}^n f(y_i | x_i, \mathbf{w}) \\ &= \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\left[\frac{(x_i^T \mathbf{w} - y_i)^2}{2\sigma^2} \right]} \end{aligned}$$

The goal is to find a $\mathbf{w}$ that maximizes the likelihood function.

$$\mathbf{w} = \arg \max_{\mathbf{w}} [L(\mathbf{w}; D)]$$

Since $\log_e(x)$ is an increasing function, we can maximize the $\log_e$ of the likelihood function.

$$\mathbf{w} = \arg \max_{\mathbf{w}} \left[\log_e (L(\mathbf{w}; D)) \right]$$

For convenience, let us denote $z(\mathbf{w}) = \log_e (L(\mathbf{w}; D))$. Let us expand $z(\mathbf{w})$

$$\begin{aligned} z(\mathbf{w}) &= \log_e \left(\prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\left[\frac{(x_i^T \mathbf{w} - y_i)^2}{2\sigma^2} \right]} \right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^n \left[\log_e \left(\frac{1}{\sigma \sqrt{2\pi}} \right) + \log_e \left(e^{-\left[\frac{(x_i^T \mathbf{w} - y_i)^2}{2\sigma^2} \right]} \right) \right] \\
&= \sum_{i=1}^n \left[\log_e \left(\frac{1}{\sigma \sqrt{2\pi}} \right) - \frac{(x_i^T \mathbf{w} - y_i)^2}{2\sigma^2} \right]
\end{aligned}$$

Note that $\log_e \left(\frac{1}{\sigma \sqrt{2\pi}} \right)$ and $2\sigma^2$ are constants. Therefore, $\mathbf{w}$ is given as

$$\mathbf{w} = \arg \max_{\mathbf{w}} \left[- \sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2 \right]$$

Instead of maximizing the above objective function, we can minimize the following

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left[\sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2 \right]$$

Notice that this is the Sum Squared Error, which we have already seen.

Conclusion

Maximizing the likelihood function for the linear regression model with **zero-mean** Gaussian noise is equivalent to minimizing the SSE (Sum Squared Error). The $\mathbf{w}$ obtained from maximizing the likelihood function given above is same as the $\mathbf{w}$ obtained from minimizing the SSE (Sum Squared Error).

6. Polynomial Regression

Let us understand polynomial regression by focusing on the following dataset.

We have n data points in $\mathbb{R}^2$. Each data point x_i has two features : $x_{(i,1)}$ and $x_{(i,2)}$. Let y_i be the label of the data point x_i , where $i \in \{1, \dots, n\}$. $y_i \in \mathbb{R}$, $\forall i \in \{1, \dots, n\}$.

Suppose, we assume that there is a cubic relationship between the features and the labels. That is, each

label is assumed to be a cubic function of the features of the corresponding data point.

$$y_i = f(x_{(i,1)}, x_{(i,2)}), \forall i \in \{1, 2, \dots, n\}$$

How would the function look like? We know that it is a cubic function. This means, the degree of every term in the function will be less than or equal to 3. Let us identify all possible terms of degree less than or equal to 3 that can be formed with $x_{(i,1)}$ and $x_{(i,2)}$.

Terms with Degree 3

$$\left(x_{(i,1)}^3\right), \left(x_{(i,2)}^3\right), \left(x_{(i,1)}x_{(i,2)}^2\right), \left(x_{(i,1)}^2x_{(i,2)}\right)$$

Terms with Degree 2

$$\left(x_{(i,1)}^2\right), \left(x_{(i,2)}^2\right), \left(x_{(i,1)}x_{(i,2)}\right)$$

Terms with Degree 1

$$\left(x_{(i,1)}\right), \left(x_{(i,2)}\right)$$

Terms with Degree 0 - An unknown constant.

The function $f(x_{(i,1)}, x_{(i,2)})$ can be described as follows:

$$f(x_{(i,1)}, x_{(i,2)}) = a + b + c + d$$

where,

$$a = \mathbf{w}_0 + \mathbf{w}_1(x_{(i,1)}) + \mathbf{w}_2(x_{(i,2)}) \quad c = \mathbf{w}_5(x_{(i,1)}x_{(i,2)}) + \mathbf{w}_6\left(x_{(i,1)}^3\right)$$

$$b = \mathbf{w}_3\left(x_{(i,1)}^2\right) + \mathbf{w}_4\left(x_{(i,2)}^2\right) \quad d = \mathbf{w}_7\left(x_{(i,2)}^3\right) + \mathbf{w}_8\left(x_{(i,1)}x_{(i,2)}^2\right) + \mathbf{w}_9\left(x_{(i,1)}^2x_{(i,2)}\right)$$

Note : a, b, c and d are used due to lack horizontal space.

Our assumption of cubic relationship between the features and the labels can be captured by the equation $X\mathbf{w} = y$, where

$$X = \begin{bmatrix} 1 & x_{(1,1)} & x_{(1,2)} & x_{(1,1)}^2 & x_{(1,2)}^2 & x_{(1,1)}x_{(1,2)} & x_{(1,1)}^3 & x_{(1,2)}^3 & x_{(1,1)}x_{(1,2)}^2 & x_{(1,1)}^2x_{(1,2)} \\ 1 & x_{(2,1)} & x_{(2,2)} & x_{(2,1)}^2 & x_{(2,2)}^2 & x_{(2,1)}x_{(2,2)} & x_{(2,1)}^3 & x_{(2,2)}^3 & x_{(2,1)}x_{(2,2)}^2 & x_{(2,1)}^2x_{(2,2)} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{(n,1)} & x_{(n,2)} & x_{(n,1)}^2 & x_{(n,2)}^2 & x_{(n,1)}x_{(n,2)} & x_{(n,1)}^3 & x_{(n,2)}^3 & x_{(n,1)}x_{(n,2)}^2 & x_{(n,1)}^2x_{(n,2)} \end{bmatrix}_{(n \times 10)}$$

$$\mathbf{w} = \begin{bmatrix} \mathbf{w}_0 \\ \mathbf{w}_1 \\ \vdots \\ \mathbf{w}_9 \end{bmatrix}_{(10 \times 1)}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{(n \times 1)}$$

- X is a $n \times 10$ matrix. Every row is a data point. Originally, each data point had two features. However, in X , each data point has 10 features. Each data point has been **transformed** from $\mathbb{R}^2$ to $\mathbb{R}^{10}$. Each column corresponds to one feature in the **transformed space**.
- y is the label vector
- $\mathbf{w}$ is the unknown vector.

Even though the features and the labels are assumed to have a cubic relationship, the equation $X\mathbf{w} = y$ is a system of linear equations with $\mathbf{w}$ as the unknown vector.

$\mathbf{w}$ is given as:

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left\| X\mathbf{w} - y \right\|_2^2$$

We also know the solution for the above optimization problem.

$$\mathbf{w} = (X^T X)^{-1} X^T y$$

7. Eigenvalues & Eigenvectors

Let matrix $A_{n \times n}$ be a square matrix and let $v \in \mathbb{R}^n$ and $v \neq 0$. Suppose,

$$Av = \lambda v$$

for some $\lambda \in \mathbb{R}$, then λ is called as an **eigenvalue** of A and $\mathbf{v}$ is called as an **eigenvector** of A with λ as the eigenvalue. Together, (λ, v) are referred to as an **eigenpair** of A . While $v \neq 0$, λ can be zero.

When A and v are multiplied, the resultant vector Av lies in the span of v . That is, $Av \in \text{Span}(v)$.

Geometrically, when A and v are multiplied,

- If $\lambda > 1$, vector v is stretched
- If $0 < \lambda < 1$, vector v is shrunk
- If $-1 < \lambda < 0$, vector v is shrunk and flipped.
- If $\lambda < -1$, vector v is stretched and flipped.

Finding Eigenvalues

Suppose $Av = \lambda v$, for some $\lambda \in \mathbb{R}$, then $\det(A - \lambda I) = 0$.

$$Av = \lambda v$$

$$Av = \lambda Iv$$

$$Av - \lambda Iv = 0$$

$$(A - \lambda I)v = 0$$

Notice that $v \in \text{Nullspace}(A - \lambda I)$. Also, $v \neq 0$. As a result, $(A - \lambda I)$ does not have a trivial nullspace. Therefore, $\det(A - \lambda I) = 0$

Solving $A - \lambda I = 0$ with λ as unknown, will give us the eigenvalues.

Example

Let A and $A - \lambda I$ be defined as follows

$$A = \begin{bmatrix} -3 & -4 & -8 \\ 4 & -1 & 8 \\ -2 & 0 & -5 \end{bmatrix}, \quad A - \lambda I = \begin{bmatrix} -3 - \lambda & -4 & -8 \\ 4 & -1 - \lambda & 8 \\ -2 & 0 & -5 - \lambda \end{bmatrix}$$

$$\begin{aligned} -2 \times \det \left(\begin{bmatrix} -4 & -8 \\ -1 - \lambda & 8 \end{bmatrix} \right) &= -2[-32 - (8 + 8\lambda)] \\ &= -2[-40 - 8\lambda] \\ &= 16\lambda + 80 \end{aligned}$$

$$0 \times \det \left(\begin{bmatrix} -3 - \lambda & -8 \\ 4 & 8 \end{bmatrix} \right) = 0$$

$$\begin{aligned}
(-5 - \lambda) \times \det \left(\begin{bmatrix} -3 - \lambda & -4 \\ 4 & -1 - \lambda \end{bmatrix} \right) &= -(5 + \lambda)[(-1)(3 + \lambda)(-1)(1 + \lambda) - (-16)] \\
&= -(5 + \lambda)[(3 + \lambda^2 + 4\lambda + 16)] \\
&= -(5 + \lambda)[\lambda^2 + 4\lambda + 19] \\
&= -(5\lambda^2 + 20\lambda + 95 + \lambda^3 + 4\lambda^2 + 19\lambda) \\
&= -\lambda^3 - 9\lambda^2 - 39\lambda - 95
\end{aligned}$$

$$\begin{aligned}
\det(A - \lambda I) &= (16\lambda + 80) - 0 + (-\lambda^3 - 9\lambda^2 - 39\lambda - 95) \\
&= -\lambda^3 - 9\lambda^2 - 23\lambda - 15
\end{aligned}$$

The determinant of $A - \lambda I$ is a cubic polynomial. This polynomial is also referred to as the **characteristic polynomial**.

$$\begin{aligned}
-\lambda^3 - 9\lambda^2 - 23\lambda - 15 &= 0 \\
-\lambda^3 - 5\lambda^2 - 4\lambda^2 - 20\lambda - 3\lambda - 15 &= 0 \\
-\lambda^2(\lambda + 5) - 4\lambda(\lambda + 5) - 3(\lambda + 5) &= 0 \\
(\lambda + 5)(-\lambda^2 - 4\lambda - 3) &= 0 \\
(\lambda + 5)(-\lambda^2 - \lambda - 3\lambda - 3) &= 0 \\
(\lambda + 5)(-\lambda(\lambda + 1) - 3(\lambda + 1)) &= 0 \\
-(\lambda + 5)(\lambda + 3)(\lambda + 1) &= 0
\end{aligned}$$

The eigenvalues are $\lambda = -1$, $\lambda = -3$ and $\lambda = -5$

Finding the Eigenvectors

We know that $(A - \lambda I)v = 0$. Therefore, $v \in \text{Nullspace}(A - \lambda I)$. For each eigenvalue λ , we have to find the nullspace of $A - \lambda I$. The basis vectors of the nullspace are the eigenvectors. Also, any non-zero linear combination of the basis vectors will also be an eigenvector with λ as the eigenvalue.

Note : The eigenvectors will be given. Readers are requested to find the nullspace and verify that the given eigenvectors are indeed correct.

Example - 1

Let us continue with matrix A in the last section.

$$A = \begin{bmatrix} -3 & -4 & -8 \\ 4 & -1 & 8 \\ -2 & 0 & -5 \end{bmatrix}$$

The eigenvalues are $-1, -3$ and -5 .

For $\lambda = -1$,

$$A - (-1)I = \begin{bmatrix} -2 & -4 & -8 \\ 4 & 0 & 8 \\ -2 & 0 & -4 \end{bmatrix}$$

Solve $(A - (-1)I)v = 0$. The vector $(-2, -1, 1)$ is a basis for the $\text{Nullspace}(A - (-1)I)$

For $\lambda = -3$

$$A - (-3)I = \begin{bmatrix} 0 & -4 & -8 \\ 4 & 2 & 8 \\ -2 & 0 & -2 \end{bmatrix}$$

Solve $(A - (-3)I)v = 0$. The vector $(-1, -2, 1)$ is a basis for the $\text{Nullspace}(A - (-3)I)$

For $\lambda = -5$

$$A - (-5)I = \begin{bmatrix} 2 & -4 & -8 \\ 4 & 4 & 8 \\ -2 & 0 & 0 \end{bmatrix}$$

Solve $(A - (-5)I)v = 0$. The vector $(0, -2, 1)$ is a basis for the $\text{Nullspace}(A - (-5)I)$

Example - 2

Consider a (3×3) identity matrix I . $Iv = v, \forall v \in \mathbb{R}^3$. Therefore, $\lambda = 1$ is the only eigenvalue of I . $(I - 1I)$ is given as

$$I - 1I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Notice that $\mathbb{R}^3$ is the null space of $(I - 1I)$. The following set is a basis set for the nullspace of $\mathbb{R}^3$.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Any non-zero linear combination of these three vectors is also an eigenvector of I with $\lambda = 1$ as the eigenvalue. Although this may seem trivial, it helps to emphasize the fact that an eigenvalue of a square matrix can have more than one eigenvector, and those vectors can be linearly independent

8. Properties of Eigenvalues and Eigenvectors

Property 1

If v is an eigenvector of A with λ as an eigenvalue, then kv is also an eigenvector of A with λ as an eigenvalue. Here, $k \in \mathbb{R}$ and $k \neq 0$

Proof

$$\begin{aligned} A(kv) &= k(Av) \\ &= k(\lambda v) \\ &= \lambda(kv) \end{aligned}$$

Any vector in $\text{Span}(v)$, except 0, is an eigenvector of A with λ as the eigenvalue.

Property 2

If u and v are two eigenvectors of A with λ as an eigenvalue, then $z = cu + dv$ is also an eigenvector of A with λ as an eigenvalue. Here, $c \in \mathbb{R}$ and $d \in \mathbb{R}$.

Proof

$$\begin{aligned} Az &= A(cu + dv) \\ &= Acu + Adv \\ &= c(Au) + d(Av) \\ &= c(\lambda u) + d(\lambda v) \end{aligned}$$

$$= \lambda(cu + dv)$$

Property 3

If v is an eigenvector of A with 0 as the eigenvalue, then $v \in \text{Nullspace}(A)$.

Proof

$$Av = 0v$$

$$= 0$$

Since $Av = 0$, $v \in \text{Nullspace}(A)$. In fact, every vector (except the zero-vector) in the null space of A is an eigenvector of A .

Property 4

The sum of all the eigenvalues of a matrix is equal to $\text{Trace}(A)$

Property 5

The product of all the eigenvalues of a matrix is equal to $\det(A)$

Proof (4 & 5)

For simplicity, we shall use a 2×2 matrix to prove the properties (4 & 5). However, the properties hold true for any square matrices.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then, $\det(A) = ad - bc$ and $\text{Trace}(A) = a + d$.

Let λ be an unknown variable. $A - \lambda I$ is given as

$$A - \lambda I = \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix}$$

We know that $\det(A - \lambda I) = 0$.

$$\det(A - \lambda I) = 0$$

$$(a - \lambda)(d - \lambda) - bc = 0$$

$$ad + \lambda^2 - (a + d)\lambda - bc = 0$$

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Notice the above characteristic polynomial. It is of the form $\lambda^2 - \lambda \times \text{Trace}(A) + \det(A)$. This is true for any 2×2 matrix.

Using quadratic formula, we get the following as the roots of the above quadratic equation.

$$\lambda_1 = \frac{(a + d) + \sqrt{a^2 + d^2 - 2ad + 4bc}}{2}$$

$$\lambda_2 = \frac{(a + d) - \sqrt{a^2 + d^2 - 2ad + 4bc}}{2}$$

Sum of the eigenvalues.

$$\begin{aligned} \lambda_1 + \lambda_2 &= \left(\frac{(a + d) + \sqrt{a^2 + d^2 - 2ad + 4bc}}{2} \right) + \left(\frac{(a + d) - \sqrt{a^2 + d^2 - 2ad + 4bc}}{2} \right) \\ &= \frac{2(a + d)}{2} \\ &= a + d \\ &= \text{Trace}(A) \end{aligned}$$

Product of the eigenvalues

$$\begin{aligned} \lambda_1 \times \lambda_2 &= \left(\frac{(a + d) + \sqrt{a^2 + d^2 - 2ad + 4bc}}{2} \right) \times \left(\frac{(a + d) - \sqrt{a^2 + d^2 - 2ad + 4bc}}{2} \right) \\ &= \left(\frac{(a + d)^2 - \left(\sqrt{a^2 + d^2 - 2ad + 4bc} \right)^2}{4} \right) \\ &= \left(\frac{a^2 + 2ad + d^2 - (a^2 + d^2 - 2ad + 4bc)}{4} \right) \\ &= \left(\frac{4(ad - bc)}{4} \right) \\ &= (ad - bc) \\ &= \det(A) \end{aligned}$$

We have used $(p^2 - q^2) = (p + q)(p - q)$ to simplify the above product.

Property 6

For a square matrix A , if $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$ and if $\lambda_1 \neq \lambda_2$, then v_1 and v_2 are linearly independent.

Proof

Let c & d be two unknown variables. The equation below has atleast one solution.

$$cv_1 + dv_2 = 0 \quad (1)$$

If $(c, d) = (0, 0)$ is the only solution, then v_1 and v_2 are linearly independent.

Multiply A on both the sides of (1)

$$\begin{aligned} A(cv_1 + dv_2) &= A0 \\ c\lambda_1 v_1 + d\lambda_2 v_2 &= 0 \end{aligned} \quad (2)$$

Consider $(1) \times \lambda_1$

$$c\lambda_1 v_1 + d\lambda_1 v_2 = 0 \quad (3)$$

Subtract $(3) - (2)$

$$\begin{aligned} d(\lambda_2 - \lambda_1)v_2 &= 0 \\ d &= 0 \end{aligned}$$

We know that $v_2 \neq 0$ as v_2 is an eigenvector. Also, $\lambda_2 \neq \lambda_1$.

Substitute $d = 0$ in (1)

$$\begin{aligned} cv_1 + 0(v_2) &= 0 \\ cv_1 &= 0 \\ c &= 0 \end{aligned}$$

Again, $v_1 \neq 0$ as v_1 is an eigenvector.

Infact, using a proof that is similar to the one given above, we can show that the eigenvectors

corresponding to distinct eigenvalues of a matrix are linearly independent.

However, if two eigenvectors are linearly independent, it does not mean that their corresponding eigenvalues are distinct. As we have seen before, for a 3×3 identity matrix, the following are the eigenvectors with 1 as the eigenvalue.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Also, the set given above is a linearly independent set.

Property 7

Let A be an $m \times n$ matrix. Then, the eigenvalues of $A^T A$ and AA^T are greater than or equal to 0.

Proof

Let $(A^T A)u = \lambda_1 u$ and let $(AA^T)v = \lambda_2 v$. Then

$$\begin{aligned} (A^T A)u &= \lambda_1 u & (AA^T)v &= \lambda_2 v \\ u^T (A^T A)u &= \lambda_1 u^T u & v^T (AA^T)v &= \lambda_2 v^T v \\ (Au)^T Au &= \lambda_1 ||u||^2 & (A^T v)^T A^T v &= \lambda_2 ||v||^2 \\ ||Au||^2 &= \lambda_1 ||u||^2 & ||A^T v||^2 &= \lambda_2 ||v||^2 \\ \lambda_1 &= \frac{||Au||^2}{||u||^2} & \lambda_2 &= \frac{||A^T v||^2}{||v||^2} \end{aligned}$$

As u and v are eigenvectors of $A^T A$ and AA^T respectively, they are non-zero vectors. Therefore, $||u||^2 \neq 0$ and $||v||^2 \neq 0$.

Also,

$$\lambda_1 = \frac{||Au||^2}{||u||^2} \geq 0 \qquad \lambda_2 = \frac{||A^T v||^2}{||v||^2} \geq 0$$

Property 8

Let A be a square matrix with λ as an eigenvalue. Then $\lambda + \hat{\lambda}$ is an eigenvalue of $A + \hat{\lambda}I$

Proof Let v be the eigenvector of A with λ as the eigenvalue. That is, $Av = \lambda v$. Then,

$$\begin{aligned}(A + \widehat{\lambda}I)v &= Av + \widehat{\lambda}Iv \\ &= \lambda v + \widehat{\lambda}v \\ &= (\lambda + \widehat{\lambda})v\end{aligned}$$

v is an eigenvector of $A + \widehat{\lambda}I$, with $\lambda + \widehat{\lambda}$ as the corresponding eigenvalue.

Property 9

Let A be a square matrix such that $Av = \lambda_1 v$ and $A^T u = \lambda_2 u$ and $\lambda_1 \neq \lambda_2$. Then, u and v are orthogonal.

Proof

$$\begin{aligned}A^T u &= \lambda_2 u \\ v^T A^T u &= \lambda_2 v^T u \\ (Av)^T u &= \lambda_2 v^T u \\ \lambda_1 v^T u &= \lambda_2 v^T u \\ (\lambda_1 - \lambda_2)v^T u &= 0 \\ v^T u &= 0\end{aligned}$$

$\lambda_1 \neq \lambda_2$, therefore, $\lambda_1 - \lambda_2 \neq 0$.

Key Idea : If v and u are eigenvectors of A and A^T respectively, corresponding to distinct eigenvalues, then v and u are orthogonal.

Point To Note

A square matrix whose entries are all real numbers can still have eigenvalues which are not real.

Example :

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Let λ be an unknown variable. Then,

$$\det(A - \lambda I) = \lambda^2 + 1$$

Solve $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

The eigenvalues are i and $-i$, where $i = \sqrt{-1}$. The number i is a **complex number**. We shall see more about complex numbers in week 5.

9. Similar Matrices

Two $n \times n$ square matrices A and B are similar if there exists an invertible $n \times n$ matrix P such that

$$A = PBP^{-1}$$

Property 1 : If A and B are similar matrices, then $\det(A) = \det(B)$

Proof :

$$\begin{aligned}\det(A) &= \det(PBP^{-1}) \\ &= \det(P) \times \det(B) \times \det(P^{-1}) \\ &= \det(B)\end{aligned}$$

We have used the fact that $\det(P) \times \det(P^{-1}) = 1$

Property 2 : If A and B are similar matrices, then $A^k = PB^kP^{-1}$ for any integer $k \geq 0$

Proof :

$$\begin{aligned}A^k &= (PBP^{-1})^k \\ &= (PBP^{-1}) \times (PBP^{-1}) \times \dots \times (PBP^{-1}) \\ &\quad \leftarrow k\text{-times} \rightarrow \\ &= PB^kP^{-1}\end{aligned}$$

Note : $P^{-1}P = I$

Property 3 : If A and B are similar matrices, then A and B have the same eigenvalues.

Proof : Let λ be an unknown variable.

$$\begin{aligned} \det(A - \lambda I) &= \det(PBP^{-1} - \lambda I) \\ &= \det(PBP^{-1} - \lambda PIP^{-1}) \\ &= \det(P(BP^{-1} - \lambda IP^{-1})) \\ &= \det(P((B - \lambda I)P^{-1})) \\ &= \det(P(B - \lambda I)P^{-1}) \\ &= \det(P) \times \det(B - \lambda I) \times \det(P^{-1}) \\ &= \det(B - \lambda I) \end{aligned}$$

$\det(A - \lambda I) = \det(B - \lambda I)$ implies that the characteristic polynomials of A and B are the same. Therefore, the roots (eigenvalues of A & B) of the polynomials will also be the same.

10. Diagonalization

An $n \times n$ square matrix A is diagonalizable if there exists an $n \times n$ diagonal matrix D such that A and D are similar. That is, there exist an $n \times n$ invertible matrix P , such that

$$A = PDP^{-1}$$

A diagonal matrix is a square matrix having 0 in all the non-diagonal entry. Suppose $a, b, c \in \mathbb{R}$ are the diagonal entries of a 3×3 diagonal matrix D , then we can denote D as follows : –

$$D = \text{Diag}(a, b, c) = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

The powers of D are easy to compute. $D^k = \text{Diag}(a^k, b^k, c^k)$, where $k \geq 0$

The determinant of a diagonal matrix is the product of all the diagonal entries.

$$\det(D) = a \times b \times c$$

The diagonal entries of D are the eigenvalues of D . Consider $D - \lambda I$, where λ is an unknown variable. We know that the solutions for $\det(D - \lambda I) = 0$ will be the eigenvalues of D .

$$\det(D - \lambda I) = 0$$

$$(a - \lambda)(b - \lambda)(c - \lambda) = 0$$

Clearly, a , b and c are the eigenvalues of D .

Since A and D are similar,

- $\det(A) = \det(D)$
- $A^k = PD^kP$
- The eigenvalues of A can be obtained from the diagonal entries of D

Let M and N be $n \times n$ square matrices such that : –

- N is a diagonal matrix
- $M = ZNZ^{-1}$, where Z is an $n \times n$ invertible matrix

The diagonal entries of N are the eigenvalues of M . Let $N = \text{Diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$. Let v_i be the i^{th} column of Z , where $i \in \{1, 2, \dots, n\}$. Then,

$$\begin{aligned}
 M &= ZNZ^{-1} \\
 MZ &= ZN \\
 M \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix} &= \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \\
 \begin{bmatrix} | & & | \\ Mv_1 & \cdots & Mv_n \\ | & & | \end{bmatrix} &= \begin{bmatrix} | & & | \\ \lambda_1 v_1 & \cdots & \lambda_n v_n \\ | & & | \end{bmatrix}
 \end{aligned}$$

We get:

$$\begin{aligned}
 Mv_1 &= \lambda_1 v_1 \\
 Mv_2 &= \lambda_2 v_2 \\
 &\vdots \\
 Mv_n &= \lambda_n v_n
 \end{aligned}$$

Therefore, $v_1, \dots, v_n$ are the eigenvectors of M .

Note : Matrix Z is not unique. This is because, kv_1 , where $k \in \mathbb{R} \setminus \{0\}$ is also an eigenvector of M , with λ_1 as eigenvalue. Therefore, we can replace the columns of Z with non-zero scalar multiples of the corresponding eigenvectors and still have

$$MZ = ZN$$

Tests For Diagonalizability

Let M be an $n \times n$ square matrix.

Test 1

- **Statement :** If the eigenvectors of M do not form a basis for $\mathbb{R}^n$, then M is not diagonalizable.
- **Reasoning :**
 - Diagonalization requires $M = ZDZ^{-1}$, where
 - Z is an $n \times n$ invertible matrix whose columns are the eigenvectors of M
 - D is a diagonal matrix containing the corresponding eigenvalues as its diagonal entries.
 - If the span of the column vectors of Z is not equal to $\mathbb{R}^n$, then Z will not be invertible

Test 2 :

- **Statement :** If the eigenvectors of M form a basis for $\mathbb{R}^n$, then M is diagonalizable.
- **Reasoning :**
 - Since the eigenvectors form a basis for $\mathbb{R}^n$, we can construct a matrix Z whose columns are the eigenvectors of M .
 - The columns of Z are linearly independent. This is because, a basis is a linearly independent set.
 - Z is invertible. Therefore, $M = ZDZ^{-1}$, where D is a diagonal matrix containing the corresponding eigenvalues as its diagonal entries.

Test 3 :

- **Statement :** If M has n distinct eigenvalues, then M is diagonalizable.
- **Reasoning :** The eigenvectors corresponding to distinct eigenvalues are linearly independent.

Caution

If M does not have n distinct eigenvalues, then we cannot say that M is not diagonalizable. For example, the identity matrix I , has 1 as the only eigenvalue. Also, $I = I \times I \times I^{-1}$. I is similar to itself. Also, I is a diagonal matrix.

Let, $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\lambda_1 \neq \lambda_2$ are the only eigenvalues of a 3×3 matrix M . Matrix M is diagonalizable if the following is true.

$$\text{nullity}(M - \lambda_1 I) + \text{nullity}(M - \lambda_2 I) = 3$$

Recall : For a square matrix M , if λ is an eigenvalue, then the vectors in a basis for the nullspace of $M - \lambda I$ are the eigenvectors of M with λ as the eigenvalue.

11. Fibonacci Sequence

A fibonacci sequence is a series of numbers, where each number is a sum of the preceeding two numbers. This sequence typically starts with 0 and 1.

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$$

Let us denote $f_0 = 0$, $f_1 = 1$, $f_2 = 1$, $f_3 = 2$ and so on. The k^{th} element of the fibonacci sequence, where k is an integer and $k \geq 2$, is given by f_k , where

$$f_k = f_{k-1} + f_{k-2}$$

One way to find the k^{th} element of the fibonacci sequence is by solving the recursive formula. As k gets larger and larger, it becomes tedious to work with the recursive formula. Diagonalization provides a clever way to find the k^{th} element of the fibonacci sequence.

We start with two linear equations.

$$f_{k+2} = f_{k+1} + f_k$$

$$f_{k+1} = f_k + f_{k-1}$$

The matrix representation of the two linear equations given above is $Au_k = u_{k+1}$, where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad u_k = \begin{bmatrix} f_{k+1} \\ f_k \end{bmatrix}, \quad u_{k+1} = \begin{bmatrix} f_{k+2} \\ f_{k+1} \end{bmatrix}$$

- $u_1 = Au_0$
- $u_2 = Au_1 = A(Au_0) = A^2u_0$
- $u_3 = Au_2 = A(A^2u_0) = A^3u_0$

In general, $u_k = A^k u_0$.

Let us find the eigenvalues and eigenvectors of A . Recall, the characteristic polynomial of a 2×2 matrix is given by

$$\det(A - \lambda I) = \lambda^2 - \lambda \times \text{Trace}(A) + \det(A)$$

- $\text{Trace}(A) = 1$
- $\det(A) = -1$

$$\det(A - \lambda I) = \lambda^2 - \lambda - 1$$

Using quadratic formula, we find that the roots of $\lambda^2 - \lambda - 1 = 0$ are λ_1 & λ_2 , where

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}, \lambda_2 = \frac{1 - \sqrt{5}}{2}$$

Let $v_1 = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$ be the eigenvector of A , corresponding to λ_1

$$(A - \lambda_1 I)v_1 = 0$$

$$\begin{bmatrix} 1 - \lambda_1 & 1 \\ 1 & -\lambda_1 \end{bmatrix} \times \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We get the following linear equations.

$$p_1 - \lambda_1 p_1 + p_2 = 0$$

$$p_1 - \lambda_2 p_2 = 0$$

Verify that $v_1 = \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix}$ is a solution for the above system of linear equations.

Similarly, we get $v_2 = \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}$ as a solution for $(A - \lambda_2 I)v_2 = 0$. Therefore, v_2 is an eigenvector of A with λ_2 as the corresponding eigenvalue.

Notice that v_1 and v_2 are linearly independent and also form a basis for $\mathbb{R}^2$. Therefore, matrix A is diagonalizable.

Let $Z = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}$ and let $D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$. The inverse of Z , is given by

$$Z^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

Now, we get $A = ZDZ^{-1}$. Therefore,

$$\begin{aligned} u_{k+1} &= A^k u_0 \\ &= (ZD^k Z^{-1}) u_0 \end{aligned}$$

Recall, $u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Let us expand $(ZD^k Z^{-1}) u_0$

$$\begin{aligned} (ZD^k Z^{-1}) u_0 &= \left(\frac{1}{\lambda_1 - \lambda_2} \right) \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \times \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \left(\frac{1}{\lambda_1 - \lambda_2} \right) \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \times \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \left(\frac{1}{\lambda_1 - \lambda_2} \right) \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} \lambda_1^k \\ -\lambda_2^k \end{bmatrix} \\ &= \left(\frac{1}{\lambda_1 - \lambda_2} \right) \begin{bmatrix} \lambda_1^{k+1} - \lambda_2^{k+1} \\ \lambda_1^k - \lambda_2^k \end{bmatrix} \end{aligned}$$

Recall, $u_k = \begin{bmatrix} f_{k+1} \\ f_k \end{bmatrix}$.

$$\begin{aligned} u_k &= (ZD^k Z^{-1}) u_0 \\ \begin{bmatrix} f_{k+1} \\ f_k \end{bmatrix} &= \left(\frac{1}{\lambda_1 - \lambda_2} \right) \begin{bmatrix} \lambda_1^{k+1} - \lambda_2^{k+1} \\ \lambda_1^k - \lambda_2^k \end{bmatrix} \end{aligned}$$

Let us simplify $f_k = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2}$.

$$\begin{aligned} (\lambda_1^k - \lambda_2^k) &= \left(\frac{1 + \sqrt{5}}{2} \right)^k - \left(\frac{1 - \sqrt{5}}{2} \right)^k \quad (\lambda_1 - \lambda_2) = \frac{(1 + \sqrt{5}) - (1 - \sqrt{5})}{2} = \frac{2\sqrt{5}}{2} = \sqrt{5} \end{aligned}$$

$$\begin{aligned} f_k &= \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2} \\ &= \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^k - \left(\frac{1 - \sqrt{5}}{2} \right)^k \right) \end{aligned}$$

For large values of k , $\left(\frac{1-\sqrt{5}}{2}\right)^k$ is approximately equal to 0.

Interesting Fact : $\frac{1+\sqrt{5}}{2}$ is referred to as the **golden ratio**.

12. Symmetric Matrices and Orthogonal Matrices

Symmetric Matrix : A square matrix S is a symmetric if $S^T = S$.

Orthogonal Matrix : A square matrix Q is an orthogonal matrix if the columns of Q form an orthonormal set.

$$Q = \begin{bmatrix} | & | & \cdots & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & \cdots & | \end{bmatrix}$$

The set $\{c_1, \dots, c_n\}$ is an orthonormal set.

Notice : $Q^T Q = I$

$$Q^T Q = \begin{bmatrix} - & c_1^T & - \\ - & c_2^T & - \\ & \vdots & \\ - & c_n^T & - \end{bmatrix} \begin{bmatrix} | & | & \cdots & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} c_1^T c_1 & c_1^T c_2 & \cdots & c_1^T c_n \\ c_2^T c_1 & c_2^T c_2 & \cdots & c_2^T c_n \\ \vdots & \vdots & \ddots & \vdots \\ c_n^T c_1 & c_n^T c_2 & \cdots & c_n^T c_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Since $Q^T Q = I$, $Q^{-1} = Q^T$. As a result, $Q Q^T = I$.

Since $Q Q^T = I$, we can say that the row vectors of Q form an orthonormal set. Let $r_1, r_2, \dots, r_n$ be the row vectors of Q

$$Q Q^T = I$$

$$\begin{bmatrix} - & r_1 & - \\ - & r_2 & - \\ & \vdots & \\ - & r_n & - \end{bmatrix} \begin{bmatrix} | & | & \cdots & | \\ r_1^T & r_2^T & \cdots & r_n^T \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$\begin{bmatrix} r_1 r_1^T & r_1 r_2^T & \cdots & r_1 r_n^T \\ r_2 r_1^T & r_2 r_2^T & \cdots & r_2 r_n^T \\ \vdots & \vdots & \ddots & \vdots \\ r_n r_1^T & r_n r_2^T & \cdots & r_n r_n^T \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

From the above equation, the following are true,

- $r_i r_i^T = 1$, for $i \in \{1, 2, \dots, n\}$
- $r_i r_j^T = 0$ for $i \in \{1, 2, \dots, n\}$ and $j \in \{1, 2, \dots, n\}$ and $i \neq j$

Therefore, the rows of an orthogonal matrix form an orthonormal set.

The following are a few facts that are true for any symmetric matrix S . The proofs for the given facts will be provided in the notes of **Week-5**.

- All the eigenvalues of a symmetric matrix are **real** numbers.
- The eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal.
- A symmetric matrix is orthogonally diagonalizable. That is, $S = QDQ^T$, where,
 - Q is an orthogonal matrix.
 - D is a diagonal matrix whose diagonal entries are the eigenvalues of S corresponding to the eigenvectors in the columns of Q .

13. Eigenvalues of Triangular Matrices and Projection Matrices

Triangular Matrix

There are two kinds of triangular matrices.

- **Upper Triangular Matrix** : An upper triangular matrix is a square matrix where all the entries below the main diagonal is 0. Following is an example of a 3×3 upper triangular matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

- **Lower Triangular Matrix** : An lower triangular matrix is a square matrix where all the entries above the main diagonal is 0. Following is an example of a 3×3 lower triangular matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix}$$

The transpose of an upper triangular matrix is a lower triangular matrix and vice-versa.

The determinant of a triangular matrix is the product of its diagonal entries. Consequently, the diagonal entries of a triangular matrix are its eigenvalues.

Projection Matrix

Let P be a projection matrix of an $m \times n$ matrix A . Matrix P project all the vectors in $\mathbb{R}^m$ onto the column space of A .

Suppose, vector $v \in C(A)$, then v is the projection of itself. That is,

$$Pv = v$$

Clearly, 1 is an eigenvalue of P . Any vector in the column space of A is an eigenvector of P with 1 as the eigenvalue.

Suppose, vector $v \in N(A^T)$, then v is orthogonal to the column space of A . The projection of v onto the column space of A is the zero vector. That is,

$$Pv = 0$$

Clearly, 0 is an eigenvalue of P . Any vector in the null space of A^T is an eigenvector of P with 0 as the eigenvalue.

Infact, 0 and 1 are the only two eigenvalues of any projection matrix. Suppose, P is an $n \times n$ projection matrix. Let v be an eigenvector of P with λ as the eigenvalue. We know that $P^2 = P$.

$$P^2 = P$$

$$P^2v = Pv$$

$$P\lambda v = \lambda v$$

$$\lambda^2 v = \lambda v$$

$$(\lambda^2 - \lambda)v = 0$$

$v \neq 0$ as it is an eigenvector. Therefore,

$$\lambda^2 - \lambda = 0$$

$$\lambda(\lambda - 1) = 0$$

$\lambda = 1$ and $\lambda = 0$ are the only two eigenvalues of P .

14. Ridge Regression

Note : In the lectures, ridge regression is covered in a minimal level. Therefore, the following will be a very brief description of ridge regression.

Overfitting :

- Assuming and learning a model that is more complex than the actual underlying model.
- For example, if we assume a quadratic relationship between the features and the labels when the actual relationship is linear, then we are overfitting

Underfitting :

- Assuming and learning a model that is less complex than the actual underlying model.
- For example, if we assume a linear relationship between the features and the labels when the actual relationship is quadratic, then we are underfitting.

In ridge regression, for $\lambda > 0$, we minimize the following objective function.

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left(\frac{1}{2} \left[\sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2 + \lambda ||\mathbf{w}||^2 \right] \right)$$

Here, λ is a **hyperparameter**. A hyperparameter is a parameter which is not learnt from the data. Instead, it is determined by trying out different values and selecting the one that produces the lowest error among all the values tried. Here, $\mathbf{w}$ is a parameter which is learnt from the data, whereas λ is a parameter which is learnt by trying out different values.

First, $\frac{1}{2}$ is included in the objective function. This is for the ease of finding the gradient of the objective function.

The objective function has two parts to it. The following represents the sum squared error, which we are already familiar with.

$$\sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2$$

The second term is $\lambda ||\mathbf{w}'||^2$. Note that $\lambda > 0$ and $||\mathbf{w}'||^2 \geq 0$. Together, $\lambda ||\mathbf{w}'||^2$ can be viewed as a penalty that is added to the sum squared error.

As we are minimizing the objective function, we are also minimizing $\lambda ||\mathbf{w}'||^2$. In ridge regression, we are trying to minimize the squared length of $\mathbf{w}$ along with the sum squared error. Higher the value of $||\mathbf{w}'||^2$, more the penalty to the loss function. In ridge regression, we are trying to learn a $\mathbf{w}$, whose norm is as low as possible.

In ridge regression, λ is referred to as the **regularization hyperparameter**. Though we shall not go into the details of λ , we shall state a few facts about it,

- Higher value of λ may lead to underfitting
- Lower value of λ may lead to overfitting

Optimization :

$$\mathbf{w} = \arg \min_{\mathbf{w}} \left(\frac{1}{2} \left[\sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2 + \lambda ||\mathbf{w}'||^2 \right] \right)$$

Let $h(\mathbf{w}) = \frac{1}{2}[p(\mathbf{w}) + q(\mathbf{w})]$ be the objective function, where

$$p(\mathbf{w}) = \sum_{i=1}^n (x_i^T \mathbf{w} - y_i)^2 \qquad q(\mathbf{w}) = \lambda ||\mathbf{w}'||^2 = \lambda \mathbf{w}^T \mathbf{w}$$

The gradient of h , is given by

$$\nabla h(\mathbf{w}) = \frac{1}{2}[\nabla p(\mathbf{w}) + \nabla q(\mathbf{w})]$$

From the discussion on linear regression, we know that

$$\nabla p(\mathbf{w}) = 2X^T X \mathbf{w} - 2X^T y$$

Gradient of $q(\mathbf{w})$ is given by:

$$\nabla q(\mathbf{w}) = 2\lambda \mathbf{w}$$

Gradient of $h(\mathbf{w})$ is given by:

$$\begin{aligned}\nabla h(\mathbf{w}) &= \frac{1}{2}[\nabla p(\mathbf{w}) + \nabla q(\mathbf{w})] \\ &= \frac{1}{2}[2X^T X \mathbf{w} - 2X^T y + 2\lambda \mathbf{w}] \\ &= X^T X \mathbf{w} - X^T y + \lambda \mathbf{w}\end{aligned}$$

Solve $h(\mathbf{w}) = 0$

$$\begin{aligned}\nabla h(\mathbf{w}) &= 0 \\ X^T X \mathbf{w} - X^T y + \lambda \mathbf{w} &= 0 \\ X^T X \mathbf{w} + \lambda \mathbf{w} &= X^T y \\ X^T X \mathbf{w} + \lambda I \mathbf{w} &= X^T y \\ (X^T X + \lambda I) \mathbf{w} &= X^T y \\ \mathbf{w} &= (X^T X + \lambda I)^{-1} X^T y\end{aligned}$$

$\mathbf{w} = (X^T X + \lambda I)^{-1} X^T y$ is a critical point. It is also the global minimum of $h(\mathbf{w})$. Again, the proof for this will be covered when we learn about convex function.

$X^T X + \lambda I$ is an invertible matrix.

- All the eigenvalues of $X^T X$ are greater than or equal to 0.
- If $\hat{\lambda}$ is an eigenvalue of $X^T X$, then $\hat{\lambda} + \lambda$ is an eigenvalue of $X^T X + \lambda I$
- $\hat{\lambda} \geq 0$
- $\lambda > 0$
- Together, $\hat{\lambda} + \lambda > 0$

All the eigenvalues of $X^T X + \lambda I$ are greater than 0. The product of all the eigenvalues of a matrix is equal to the determinant of that matrix. Clearly, $\det(X^T X + \lambda I) > 0$. Therefore, $(X^T X + \lambda I)$ is invertible.
